

LOVELY PROFESSIONAL UNIVERSITY



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P ROFESSIONAL
U NIVERSITY

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INTRODUCTION

User Interface (UI) plays a critical role in shaping how users perceive, understand, and interact with shared physical and digital systems within large institutional environments. On a university campus, an effective UI is essential not only for aesthetic appeal but also for clarity, accessibility, and efficient decision-making. When UI elements fail to communicate essential information, the overall user experience (UX) deteriorates, affecting convenience, confidence, and inclusivity for diverse user groups.

In a large and dynamic campus such as Lovely Professional University (LPU), thousands of students, faculty members, staff, and visitors navigate multiple academic blocks and facilities every day. These users rely on clear visual cues, signage, and system updates to move efficiently—especially when facilities such as elevators, staircases, or key services are unavailable or congested. However, the absence of a well-designed UI framework creates significant challenges in daily navigation and planning.

One of the major issues is the lack of real-time status displays that inform users about crucial conditions such as maintenance activities, temporary shutdowns, or overcrowding of facilities. Without timely and visible updates, users are left uncertain, often discovering issues only after reaching the location. This leads to wasted time, confusion, and stress—particularly during time-sensitive situations such as attending lectures, examinations, or administrative appointments.

Additionally, limited and poorly placed signage in blocks without elevators further exacerbates accessibility concerns. New students, visitors, and individuals with temporary or permanent mobility limitations struggle to find alternative routes or facilities due to unclear navigation cues. The absence of inclusive signage reduces autonomy and forces users to depend on others for guidance, negatively impacting their sense of independence and comfort within the campus environment.

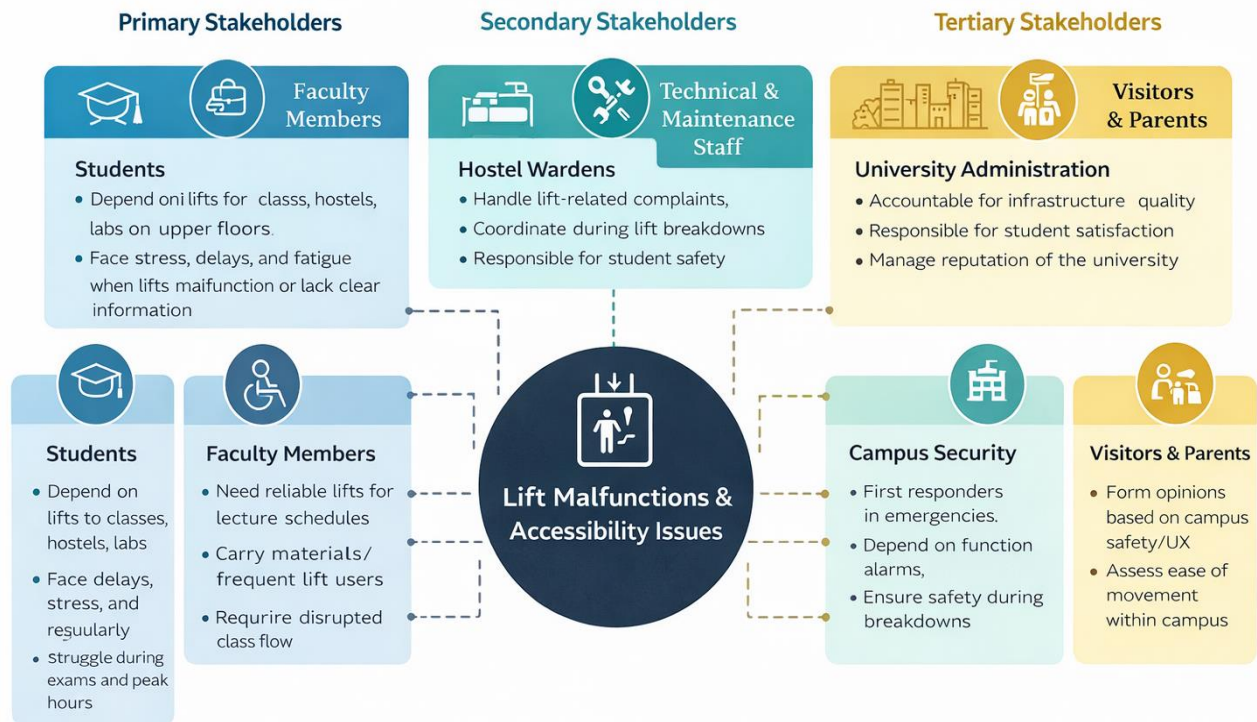
From a UX perspective, these UI and accessibility shortcomings undermine usability, efficiency, and trust in campus infrastructure. They contribute to frustration, reduced productivity, and an overall negative perception of institutional management. This case study examines the UI-related challenges arising from the lack of real-time information systems and inadequate signage, analysing their impact on user behaviour, accessibility, and satisfaction. By applying user-centered design principles, the study aims to propose solutions that enhance clarity, inclusivity, and confidence—demonstrating that effective UI design is fundamental to creating a seamless and supportive campus experience.

STAKEHOLDERS

Stakeholder Analysis

Lift Malfunction & Poor Accessibility Experience on Campus

Understand the key stakeholders impacted by lift issues to identify pain points and opportunities for repvement.



Stakeholders are individuals or groups who are directly or indirectly affected by the functioning and reliability of campus lifts. Since lifts are essential for movement across multi-level buildings, identifying stakeholders helps understand different user needs, safety concerns, accessibility requirements, and expectations for smooth campus mobility. A well-maintained lift system contributes to improved punctuality, reduced physical strain, and a more positive campus experience. The lift problem affects different user groups on campus who interact with or manage the lift system in various ways. Stakeholders include both direct users who rely on lifts for daily movement and indirect users responsible for maintenance, safety, and decision-making. Understanding their needs, frustrations, and influence is important to develop a reliable and safe lift experience for everyone at LPU.

Three types of Stakeholders are defined here:

1. Primary Stakeholders
2. Secondary Stakeholders
3. Tertiary Stakeholders

Issue: Lift Malfunction and Poor Accessibility Experience on Campus

The primary stakeholders are those who interact with the lift system daily and are most directly affected by its malfunctioning. Students rely heavily on lifts to access classrooms, hostels, and laboratories located on upper floors. When lifts fail or lack clear information, students experience stress, delays, and physical fatigue, especially during peak hours, examinations, and tightly scheduled lectures. Faculty members are also significantly impacted, as they frequently move across floors carrying teaching materials and equipment. Lift delays disrupt lecture schedules, affect class flow, and reduce overall teaching efficiency. Individuals with mobility challenges—including people with disabilities, temporary injuries, or health conditions—are among the most vulnerable users. For them, malfunctioning lifts or poor guidance systems reduce independence, create safety risks, and cause emotional discomfort due to unclear signage and unreliable accessibility support.

Secondary stakeholders include operational and support personnel who manage the functioning and safety of the lift system. Hostel wardens often serve as the first point of contact for students reporting lift-related issues and are responsible for coordinating responses during breakdowns or emergencies. The lack of transparency and real-time system updates increases their workload and operational pressure. Technical and maintenance staff are tasked with repairs and regular upkeep; however, the absence of real-time reporting mechanisms leads to delayed responses and forces a reactive rather than preventive maintenance approach. Campus security personnel act as first responders during lift failures or emergencies and depend on properly functioning emergency buttons and alert systems. When communication tools fail, their ability to ensure user safety is compromised.

Tertiary stakeholders consist of decision-makers and groups that influence long-term planning and public perception of the university. The university administration is accountable for maintaining infrastructure quality and ensuring safety standards. Poor lift usability and accessibility directly affect student satisfaction and damage the institution's reputation, highlighting the need for data-driven infrastructure improvements. Construction and planning authorities play a critical role in designing building layouts and vertical transport systems; however, insufficient user feedback often results in long-term accessibility gaps. Visitors and parents also form an important stakeholder group, as they judge campus quality based on basic infrastructure reliability. Confusion, inconvenience, and poor signage during their visits create negative impressions, making UI and wayfinding systems crucial first touchpoints of the overall campus experience.

USER RESEARCH – QUESTIONNAIRE

User research plays a vital role in gaining an in-depth understanding of the experiences, challenges, and expectations of campus users impacted by aging infrastructure. To collect qualitative data, structured interviews were conducted using a carefully designed questionnaire that examined perceptions related to safety, accessibility, maintenance, and the overall campus environment.

Followings are the questions being asked in interviews:

1. How frequently do you use lifts on campus, and for what purposes (classes, hostel, work, etc.)?
2. Have you experienced lift malfunctions on campus? If yes, how often do they occur?
3. What types of issues have you faced most commonly (e.g., non-working buttons, sudden stops, slow movement)?
4. How do lift breakdowns affect your daily schedule or punctuality?
5. Have lift issues ever caused you to miss or be late for a class, meeting, or exam?
6. How safe do you feel while using campus lifts, especially during malfunctions?
7. Are emergency features such as alarm or help buttons clearly visible and functional?
8. How easy is it to understand whether a lift is working, under maintenance, or overcrowded?
9. Do you find the signage and directions helpful in buildings without elevators?
10. How do lift problems affect individuals with mobility challenges or temporary injuries?
11. What emotions do you usually experience when a lift is not working (stress, frustration, anxiety, etc.)?
12. How do you currently report lift-related problems, if at all?
13. Do you feel that maintenance responses are timely and effective?
14. What improvements would you like to see in lift design, information display, or signage?
15. Overall, how satisfied are you with the lift system on campus, and why?

INTERVIEWS

To gain deeper insights into user experiences with the campus lift system, interviews were conducted with five individuals who regularly rely on lifts for daily movement. The discussions explored key aspects such as system reliability, safety, accessibility, and the effect of lift-related issues on routine campus activities. The objective was to understand how recurring breakdowns, unresponsive controls, and delays in maintenance disrupt academic schedules and operational efficiency. This section presents brief profiles of the participants along with the major observations and insights gathered from the interviews.

1. Interview 1

Name: Vaibhav Singh

Role: Student

Year: 3rd year

Program: B tech- Computer Science and Engineering

Institution: Lovely Professional University



Q1. How frequently do you use lifts on campus, and for what purposes (classes, hostel, work, etc.)?

Answer:

I use the lifts on campus only occasionally, mainly when I am carrying heavy items or feeling unwell. Most of the time, I prefer using the stairs, so lift usage is not a regular part of my daily routine.

Q2. Have you experienced lift malfunctions on campus? If yes, how often do they occur?

Answer:

Yes, I have experienced lift malfunctions on campus. They occur quite frequently—around 2–3 times a week, especially during peak hours in the morning and afternoon.

Interview Insights:

- Buttons and display malfunction cause confusion and uncertainty about whether to wait or leave.
- Lift misbehavior creates **fear and safety anxiety**, especially since emergency features don't work.
- Visitors rely completely on campus infrastructure → negative experience affects perception of university quality.

Q3. What types of issues have you faced most commonly (e.g., non-working buttons, sudden stops, slow movement)?

Answer:

"Yes, I have faced multiple issues while using lifts on campus. The most common problem is that the lift buttons often do not respond immediately, especially during peak hours. At times, the lift moves very slowly or stops for longer than expected on each floor. I have also experienced situations where the display screen was not working, making it difficult to know the current floor or direction."

2. Interview 2

Name: Neeraj

Role: Staff in placement cell

Institution: Lovely Professional University



Q1. How do lift breakdowns affect your daily schedule or punctuality?

Answer:

Yes, many times. When the lift doesn't work properly before class, I'm forced to take the stairs and end up reaching late. This becomes a bigger issue when teachers mark attendance strictly. It adds unnecessary stress to my day.

Q2. Have lift issues ever caused you to miss or be late for a class, meeting, or exam?

Answer:

Yes, lift issues have caused delays multiple times. On several occasions, lifts stopped unexpectedly or took unusually long to arrive, which made me late for lectures and one internal assessment. During

peak hours, waiting for a working lift often disrupted my schedule and increased stress, especially when classes were on higher floors.

Interview Insights:

- Lift issues affect **punctuality**, leading to stress and late attendance.
- Safety concerns arise due to sudden jerks and inactive emergency features, especially when alone.
- Students feel anxious and **lack trust** in lift reliability.

3. Interview 3

Name: Rohit Singh

Role: Student

Year: 3rd year

Program: B tech- Computer Science and Engineering

Institution: Lovely Professional University



Q1. Are emergency features such as alarm or help buttons clearly visible and functional?

Answer:

Most of the time, the emergency features are either not clearly visible or do not inspire confidence. While the alarm or help button is present, users are unsure whether it actually works, as there is no clear indication, feedback, or instruction displayed. This creates anxiety, especially during lift breakdowns or unexpected stops.

Q2. How easy is it to understand whether a lift is working, under maintenance, or overcrowded?

Answer:

It is generally difficult to understand the status of the lift before using it. There is no clear indicator or display showing whether the lift is operational, under maintenance, or overcrowded. Most of the time, users realize the issue only after waiting for several minutes or reaching the lift area.

Interview Insights:

- Lack of real-time status information causes confusion and unnecessary waiting.

- Users often waste time reaching non-functional lifts due to absence of signage.
- Poor status visibility increases frustration, especially during peak hours or urgent situations.

4. Interview 4

Name: Amit Kumar

Role: Technician

Institution: Lovely Professional University



Q1. What are the main technical reasons behind the lift buttons and display not working properly, and how often do these issues occur?

Answer:

Most of the time, the button malfunctions are caused by wiring faults or excessive force used by students which damages the internal switches. The display problems occur due to loose connectors or outdated control modules. These issues happen very frequently, especially during peak hours when the lift is continuously used. Some parts are old and need replacement, but we are waiting for approvals from the administration.

Q2. What difficulties do you face while repairing the lifts, and what support or resources would help you fix the issues faster?

Answer:

The biggest challenge is the delay in receiving spare parts and official approvals. We perform temporary resets or quick fixes, but without proper replacements, the same issues come back repeatedly. We also handle many complaints at once, so managing time becomes difficult. If we had better tools, monitoring systems, and timely administrative support, we could prevent breakdowns instead of repairing them again and again.

Interview Insights:

- Hardware issues occur frequently due to **wiring damage** and outdated components.
- Regular breakdowns result from delayed approvals and lack of spare parts.
- Technicians feel pressured due to **repetitive temporary fixes** instead of long-term solutions.

5. Interview 5

Name: Sonu Kumar

Role: Administrative Staff

Institution: Lovely Professional University



Q1. What improvements would make the lift experience better?

Answer:

Proper replacement of faulty buttons, fixing the display system, and making sure the emergency buttons work are essential. Also, quicker maintenance responses and a digital indicator of lift status on each floor would help a lot.

Q2. How would you rate the overall lift experience at LPU and why?

Answer:

I would rate it 2 out of 5. The lift is supposed to make access easier, but right now it causes more inconvenience and delay. With proper working condition and regular maintenance, it could greatly improve daily campus movement.

Interview Insights:

- Acknowledges that current lift condition causes more inconvenience than convenience.
- Identifies clear technical improvements needed (buttons, display, emergency function).
- Low satisfaction rating shows urgency for improved maintenance workflow.

USER PERSONA



Vaibhav Singh

21 years old

3rd Year B.Tech – Computer Science & Engineering

Lovely Professional University

“I usually take the stairs, but I use the lift when I'm carrying heavy stuff or not feeling well.”

GOALS	FRUSTRATIONS	BEHAVIORS	PAIN POINTS
- Get to class and hostel efficiently - Stay safe on campus - Focus on studies without hassle	- Frequent lift malfunctions - Unresponsive buttons - Fear of getting stuck in the lift	- Prefers using stairs over the lift - Uses lift occasionally when carrying heavy items - Avoids lifts during busy hours	- Buttons not working properly - Lift moving slowly or stopping for long - Display screen not showing floor info

INSIGHTS

CONFUSION	SAFETY CONCERNS	VISITOR EXPERIENCE
Malfunctioning buttons and displays make it unclear whether to wait or leave.	- Frequent breakdowns lead to anxiety and fear - Emergency features often don't work	- Visitors rely entirely on campus infrastructure - Frequent lift issues give a poor impression of the university.



Neeraj

34 years old

Staff in Placement Cell

Lovely Professional University

“Lift breakdowns often make me late, adding unnecessary stress to my already busy day.”

GOALS	FRUSTRATIONS	BEHAVIORS	PAIN POINTS
- Arrive on time for classes, meetings, and exams - Manage a busy schedule efficiently - Reduce stress from daily disruptions	- Frequently late because of lift breakdowns - Stress from waiting for malfunctioning lifts - Missing or being late for important commitments	- Avoids lifts during peak hours if possible - Uses lifts regularly to save time and energy - Takes the stairs reluctantly when lift is broken	- Lifts frequently break down during peak hours - Waiting too long for a working lift - Missing the start of classes and meetings

INSIGHTS

PUNCTUALITY ISSUES	STRESS & PRESSURE	WORKPLACE PERCEPTION
- Lift failures often cause lateness for classes and meetings - Delays lead to stress about missing important commitments	- Frequent lift waiting increases stress and frustration - Running late adds pressure to the already tight schedule - Higher stress levels due to unpredictability of lifts	- Staff reliability is affected by lift issues - Frequent lift malfunctions give a poor impression of the university's management.



Rohit Singh

20 years old

3rd Year B.Tech - Computer Science & Engineering

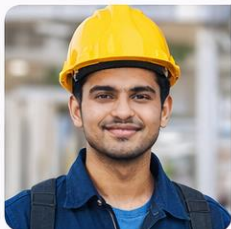
Lovely Professional University

“ I'm never sure if the emergency buttons actually work. It's stressful when the lift suddenly stops. ”

GOALS	FRUSTRATIONS	BEHAVIORS	PAIN POINTS
• Feel safe and secure while using the lift • Understand the lift's status before entering • Have quick and reliable help in emergencies	• Unclear if emergency buttons and alarms work • No clear indication if the lift is functioning • Anxiety during lift breakdowns due to low confidence in help features	• Nervous about using the lift during busy times • Stresses when lift suddenly stops • Avoids using crowded or potentially faulty lifts	• Emergency buttons not clearly marked • Lack of confidence in emergency features • No indications if lift is under maintenance or crowded

INSIGHTS

CONFUSION	SAFETY CONCERNS	TIME WASTE
• No real-time status info causes uncertainty • Users waste time waiting for out-of-service lifts	• Anxiety from unreliable emergency help • Lack of feedback on alarm and help buttons	• Wasted efforts reaching nonfunctional lifts • Long wait times to find a working lift



Amit Kumar

35 years old

Technician

Lovely Professional University

“ The lift issues happen mostly during peak hours. We're waiting on parts approval and better tools to prevent constant breakdowns. ”

GOALS	FRUSTRATIONS	CHALLENGES	WORKING CONDITIONS
• Ensure lifts are operational and safe • Get parts and approvals quickly • Prevent frequent breakdowns with better tools	• Electrical issues like wiring faults and worn switches • Display problems due to loose connectors and outdated • Frequent breakdowns due to overdue replacements	• Long delays waiting for spare parts and approvals • Performing temporary fixes instead of permanent repairs • Handling multiple complaints at once, causing time pressure	• Outdated control modules need replacement • Temporary resets don't solve root problems • Approval process is slow, causing repair delays

INSIGHTS

SPARE PART DELAYS	RESOURCE LIMITATIONS	TIME MANAGEMENT
• Repairs delayed due to long waits for spare parts • Admin approvals slow down entire repair process	• Lacking modern tools and monitoring systems for efficient fixes • Performing quick fixes due to lack of proper replacements	• Multiple complaints make it hard to manage time • Preventive maintenance is difficult without better resources



Sonu Kumar

40 years old

Administrative Staff

Lovely Professional University

“ The lifts should improve access, not cause delays.
Faster responses and clear indicators of lift status
would help a lot. ”

GOALS	SUGGESTIONS	CONCERNS	
- Enhance lift reliability and safety - Improve staff and student satisfaction - Reduce delays and complaints about lifts	- Replace faulty buttons with quality parts - Repair and update the display system - Ensure emergency buttons always function	- Frequent complaints about lift breakdowns - Administrative pressure due to ongoing issues - Current lift status causes dissatisfaction among staff and students	★★★★★ > ★★★★★ I would rate it 2 out of 5. The lifts cause more inconvenience right now. With better maintenance and updates, they could greatly improve daily campus movement.

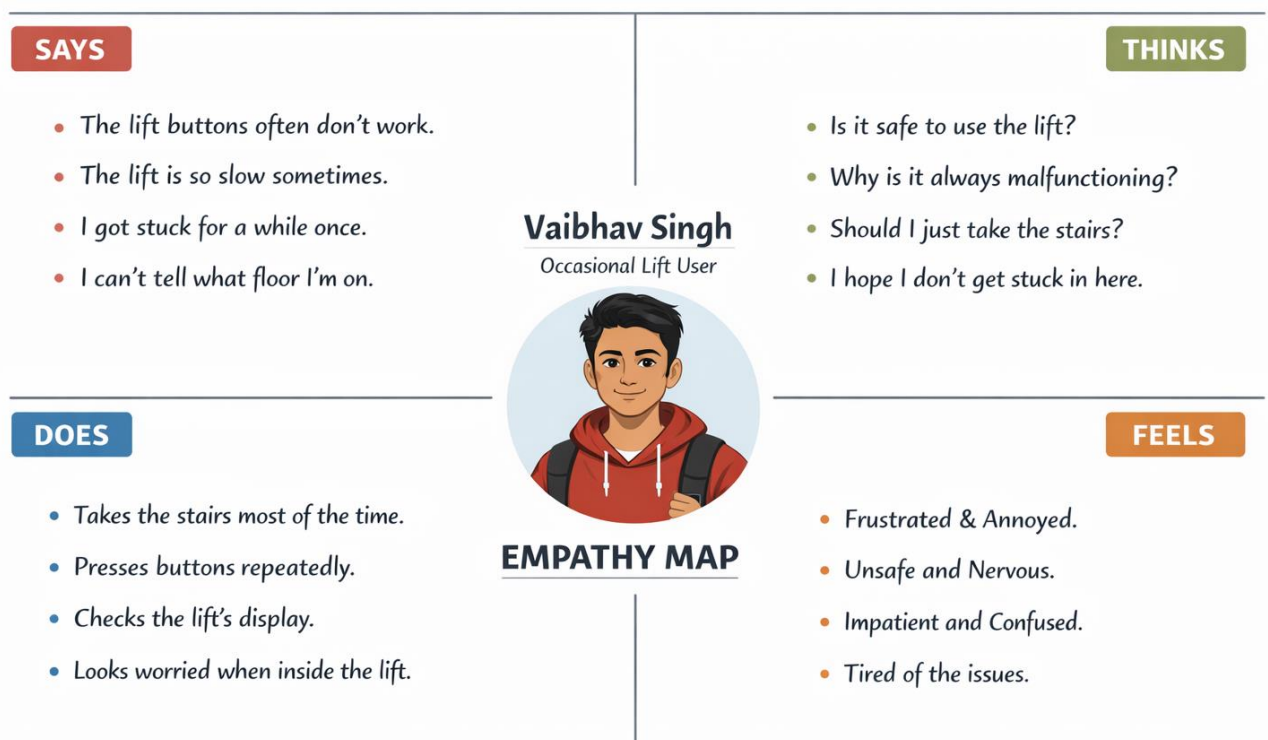
INSIGHTS

SATISFACTION	RELIABILITY	TRANSPARENCY
- Frequent breakdowns frustrate both staff and students - Improved lift reliability would raise satisfaction ratings	- Quicker repairs would minimize downtime - Better parts and equipment prevent recurring faults	- Digital indicators would provide clear, real-time lift status - Knowing lift status before waiting would reduce frustration

EMPATHY MAPPING

Empathy mapping is a UX method used to understand what users think, feel, say, and do when interacting with the campus lifts. It helps identify their emotions, frustrations, expectations, and behavior while using the lift system. These empathy maps highlight how lift issues affect students' punctuality, comfort, and confidence, while also revealing the pressure faced by technicians and administration in maintaining efficient lift operations.

1. Empathy Map 1: Vaibhav: Third Year Student



2. Empathy Map 2: Neeraj: Administrative Staff

SAYS

- The lift was broken again today.
- I was late because the lift took forever.
- I had to take the stairs up to the 7th floor.
- It's really stressful when this happens.

Neeraj

Staff in Placement Cell
at Lovely Professional University



THINKS

- Why are these lifts so unreliable?
- Will I be late to class or meeting?
- Is it safe to be in this lift alone?
- I hope I don't get stuck before a class.

DOES

- Takes the stairs if lift is broken.
- Arrives late to meetings or classes.
- Waits impatiently for the lift.
- Checks the time frequently.

EMPATHY MAP

FEELS

- Stressed and Anxious.
- Frustrated & Impatient.
- Unsafe when lift jerks suddenly.
- Worried about frequent delays.

SOLUTION REPORT:

Improving Lift Reliability & User Experience at LPU

Problem Overview:

LPU campus lifts frequently malfunction due to non-responsive buttons, faulty floor displays, and non-working safety features like the NS (Non-Stop) button. These issues cause delays, anxiety, class disruptions, and negative perceptions of campus infrastructure.

Both primary users (students & faculty) and secondary users (technicians & administrative staff) experience difficulties, although in different ways. This solution aims to improve lift safety, efficiency, and usability for all stakeholders.

Key User Insights:

- Students feel anxious, unsafe, and frustrated when lifts fail during peak hours.
- Repeated issues impact punctuality and increase physical strain when switching to stairs.
- Technicians face continuous pressure and depend on approvals/resources for repairs.
- Administration wants to reduce complaints but lacks real-time system visibility.

Smart Lift Monitoring & Maintenance System — Solution Components

1. Durable Lift Buttons & Display Replacement

The current lift buttons frequently fail to respond and the display screens often remain blank or flicker. By replacing them with high-quality and long-lasting hardware, lift performance becomes smooth and reliable. This will increase user confidence and reduce frustration during peak hours.

2. Real-Time Floor Status Display

A digital status screen will be installed both inside and outside the lift to show its exact floor position and direction of movement. This helps users understand whether the lift is coming or already full, reducing confusion, waiting time, and overcrowding around the lift area.

3. Emergency/NS Button Alert System

The emergency/NS button will be made fully functional and connected to an alert mechanism that directly notifies technicians. In case the lift gets stuck or stops unexpectedly, technicians can respond immediately, ensuring better safety and reducing user panic.

4. Mobile Maintenance Dashboard

Technicians will receive live notifications whenever an issue occurs. This dashboard will show the fault type, lift location, and repair status, allowing technicians to fix problems faster and prevent repeated system breakdowns through predictive maintenance.

5. QR-Based Issue Reporting System

A QR code placed inside the lift will allow students and staff to report any malfunction instantly using their phones. This eliminates communication delays and gives precise information about the malfunctioning lift, resulting in quicker troubleshooting and transparent complaint tracking.

6. Maintenance Scheduling Tracker

Instead of waiting for the lift to fail, scheduled reminders will ensure timely inspections and servicing. This proactive approach helps prevent major failures, reducing repetitive breakdowns and improving long-term reliability of the lift system.

How the Solution Solves the Problem

1. Buttons Not Working → Durable Hardware Replacement → Smooth & Quick Operation

The main issue faced by users is that lift buttons do not respond, causing delays and frustration. By replacing them with durable and high-quality hardware, the lift will react instantly to button presses and reduce the time users waste waiting for the lift to register their input.

2. No Display or Location Information → Real-Time Digital Floor Indicators → Clear Navigation & Safety

When the display screen is blank, users don't know whether the lift is coming or already full, which creates confusion and anxiety. Installing digital floor status screens will allow users to see the exact position and direction of the lift, improving navigation and ensuring a safer and more organized waiting experience.

3. Slow Maintenance Response → Technician Alert System → Faster Repairs, Fewer Complaints

Curre

ntly, technicians get informed late about lift failures, leading to delays in fixing issues. The new alert system will notify technicians instantly when a fault occurs, so they can respond faster and reduce the number of repeated complaints about the same issue.

4. Users Feel Unsafe → Emergency Alert Activation → Increased Trust & Safety

Because safety buttons often fail, users fear getting stuck inside the lift. A functioning emergency alert system will directly notify technicians, ensuring immediate support during breakdowns. This significantly improves user confidence in lift safety.

5. Complaints Ignored → QR-Based Issue Reporting & Tracking → Transparent Maintenance Process

Many users feel that even after reporting problems, no visible action is taken. With a QR-based reporting feature, every complaint will be recorded and tracked, making the repair process more transparent and accountable — users will know their concern is taken seriously.

User-Centered Design Justification

Fear of getting stuck

Students expressed anxiety when the lift stops unexpectedly or the emergency button doesn't work.

→ This insight led to implementing a fully functional emergency alert system to increase user safety and confidence.

Delays causing late arrival

Users shared that unreliable lifts result in missed attendance and rushed mornings.

→ This inspired improvements in system speed and real-time visibility, helping students reach classrooms on time.

Lack of communication

Users often feel confused when the lift location isn't visible or when it stays idle without indication.

→ To solve this, digital floor indicators and QR reporting were added for clarity and better information flow.

Multiple complaints with no action

Administration and technicians feel pressured due to repeated breakdown reports.

→ Smart maintenance tools and alert dashboards ensure faster response and reduced repetitive issues.

Confusion → Awareness Fear → Safety

Delay → Efficiency

CONCLUSION

The analysis of the lift system at Lovely Professional University reveals that frequent malfunctions such as unresponsive buttons, faulty display screens, and non-functional emergency features significantly affect the daily experience of campus users. These issues not only cause delays and inconvenience but also create emotional stress, discomfort, and safety concerns for students, faculty, visitors, and individuals with mobility challenges. Maintenance staff, on the other hand, face pressure due to a lack of timely resources and repeated complaints without permanent solutions.

By applying user-centered design methods — including stakeholder analysis, interviews, personas, and empathy mapping — the project identified the real needs and expectations of users. The proposed Smart Lift Monitoring & Maintenance System offers a comprehensive solution that improves hardware reliability, ensures rapid fault response, increases transparency in issue reporting, and enhances safety within lift environments.

Implementing this solution will result in a more reliable lift experience, improved punctuality, and higher satisfaction among users, while supporting maintenance staff with efficient tools. Ultimately, the project demonstrates that improving lift infrastructure is not merely a technical upgrade, but a crucial UX enhancement that promotes accessibility, confidence, and a positive perception of the university campus.

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